

Accumulation Functions and the Fundamental Theorem Answer Key

AP Calculus AB/BC

Quick Answers

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|---|---|
| 1. 79 | 5. (a) $C'(4) = -4$, — |
| 2. 145 L | 6. (a) $V'(8) = -2$, (b) x where V peaks = 6, — |
| 3. (a) $P'(3) = 3$, — | 7. (a) $V(6) = 18 \text{ m}^3$, (b) $V(10) = 10 \text{ m}^3$ |
| 4. (a) exact $Q'(2) = 4\sqrt{65}$, (b) $Q'(2)$ to two decimals = 32.25 | 8. (a) exact $F'(1) = \frac{2}{e} - \frac{2}{e^4}$, (b) $F'(1)$ to three decimals = 0.699, — |
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What is verified

12 of 16 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6 – *problems 1, 3, 4, 5, 6, 8*

LIM-5 – *problems 2, 7*

Difficulty 1–5 of 5 across 16 tagged checks.

- 1.** Tank, $r(t) = 3t^2 + 4 \text{ L/min}$, $V(t) = \int_0^t (3u^2 + 4) du$. Find $V'(5)$.

Solution. The Fundamental Theorem says the derivative of an accumulation function is the integrand evaluated at the upper limit — no antiderivative required: $V'(t) = 3t^2 + 4$, so $V'(5) = 3(25) + 4$.

$$\boxed{V'(5) = 79}$$

Litres per minute, which is simply the pump's rate at that moment.

- 2.** Same tank. Find the volume added during the first 5 minutes.

Solution. Here we do need an antiderivative: $\int (3t^2 + 4) dt = t^3 + 4t$, so

$$V(5) = [t^3 + 4t]_0^5 = (125 + 20) - 0.$$

$$\boxed{145 \text{ L}}$$

3. $P(x) = 200 + \int_0^x \frac{12}{1+t} dt$ thousand cells. (a) $P'(3)$. (b) Why the culture never shrinks.

Solution (a). The constant 200 differentiates away and the integral contributes its integrand at x : $P'(x) = \frac{12}{1+x}$, so $P'(3) = \frac{12}{4}$.

$$\boxed{P'(3) = 3}$$

Thousand cells per day.

Solution (b) — instructor-judged. $P'(x) = \frac{12}{1+x}$ is positive for every x in $[0, 10]$, and a function whose derivative is positive throughout an interval is increasing on it. Credit any wording that connects the sign of the integrand to the sign of P' ; a response that only says “the integral keeps adding” without naming the derivative has not used the theorem.

4. $Q(x) = \int_1^{x^2} \sqrt{1+t^3} dt$. (a) Exact $Q'(2)$. (b) To two decimals.

Solution (a). The upper limit is x^2 , not x , so the chain rule attaches its derivative:

$$Q'(x) = \sqrt{1+(x^2)^3} \cdot 2x = 2x\sqrt{1+x^6}.$$

At $x = 2$: $\sqrt{1+64} = \sqrt{65}$ and $2x = 4$.

$$\boxed{Q'(2) = 4\sqrt{65}}$$

Solution (b).

$$\boxed{Q'(2) \approx 32.25}$$

Millicoulombs per second.

5. $C(x) = \int_x^{12} (0.5t + 2) dt$ litres. (a) $C'(4)$. (b) Why C' is negative.

Solution (a). Swap the limits to put the variable on top: $C(x) = -\int_{12}^x (0.5t + 2) dt$, so $C'(x) = -(0.5x + 2)$ and $C'(4) = -(2 + 2)$.

$$\boxed{C'(4) = -4}$$

Litres per hour.

Solution (b) — instructor-judged. The variable is the *lower* limit. Reversing the limits to apply the theorem introduces the minus sign, and the physical reading agrees: raising the lower limit throws away coolant that had already been counted, so the amount remaining falls. Credit either the limit-reversal argument or the physical one.

6. Pond, $r(t) = 6 - t$ m³/min, $V(x) = \int_0^x (6 - t) dt$. (a) $V'(8)$. (b) Where V peaks. (c) Justify.

Solution (a). $V'(x) = 6 - x$, so $V'(8) = 6 - 8$.

$$\boxed{V'(8) = -2}$$

Solution (b). $V' = 0$ when $6 - x = 0$.

$$\boxed{x = 6}$$

Solution (c) — instructor-judged. $V'(x) = 6 - x$ is positive for $x < 6$ and negative for $x > 6$, so V rises up to $x = 6$ and falls after it; that makes $x = 6$ the absolute maximum on $[0, 10]$. Full credit needs the sign change of the rate. “ $V'(6) = 0$ ” alone locates a candidate, not a maximum.

7. Same pond. (a) $V(6)$. (b) $V(10)$.

Solution (a). $\int_0^6 (6 - t) dt = \left[6t - \frac{t^2}{2} \right]_0^6 = 36 - 18$.

$$\boxed{V(6) = 18 \text{ m}^3}$$

Solution (b). $\int_0^{10} (6 - t) dt = 60 - 50$.

$$\boxed{V(10) = 10 \text{ m}^3}$$

The pond gains water for six minutes and then gives most of it back, which is exactly what the sign change in part 6 predicted.

8. $F(x) = \int_{2x}^{x^2} e^{-t^2} dt$. (a) Exact $F'(1)$. (b) To three decimals. (c) Why F' is not $e^{-x^4} - e^{-4x^2}$.

Solution (a). Split at a constant, say 0:

$$F(x) = \int_0^{x^2} e^{-t^2} dt - \int_0^{2x} e^{-t^2} dt.$$

Each piece contributes its integrand at its own limit, times that limit's derivative:

$$F'(x) = e^{-(x^2)^2} \cdot 2x - e^{-(2x)^2} \cdot 2 = 2x e^{-x^4} - 2e^{-4x^2}.$$

At $x = 1$ this is

$$\boxed{F'(1) = 2e^{-1} - 2e^{-4}}$$

Solution (b).

$$\boxed{F'(1) \approx 0.699}$$

Solution (c) — instructor-judged. Both limits move, so both contribute, and each contribution carries the derivative of its own limit — $2x$ from the upper limit and 2 from the lower. Dropping those factors is the error the shortcut form makes. Full credit requires naming both chain-rule factors, not merely observing that there is a subtraction.